

Question	Answer	Marks
4(a)(i)	product of force and <u>perpendicular</u> distance (between the line of action of the force and the point)	B1
4(a)(ii)	(the line of action of) the <u>weight</u> passes through P / the perpendicular distance of (the line of action of) the <u>weight</u> from P is zero (so the moment is zero)	B1
4(a)(iii)	$2T \sin \theta = W$	C1
	$\sin \theta = 19 / (2 \times 13)$	A1
	$\theta = 47^\circ$	
4(a)(iv)	$E = \sigma / \varepsilon$ or $E = FL / Ax$ and $\varepsilon = x / L$	C1
	$\varepsilon = F / (AE)$	
	$= 13 / (\pi (0.8 \times 10^{-3} / 2)^2 \times 9.3 \times 10^{10})$	C1
	$= 2.8 \times 10^{-4}$	A1

Question	Answer	Marks
4(b)(i)	$s = ut + \frac{1}{2} at^2$ $1.6 = \frac{1}{2} \times 9.81 \times t^2$	C1
	$t = ((2 \times 1.6) / 9.81)^{0.5}$ $= 0.57 \text{ s}$	A1
	OR $v = (2 \times 9.81 \times 1.6)^{0.5} = 5.6$ $t = (v - u) / a$ $= (5.6 - 0) / 9.81$	(C1)
	$= 0.57 \text{ s}$	(A1)
	OR $t = 2s / v$ $= 2 \times 1.6 / 5.6$	(C1)
	$= 0.57 \text{ s}$	(A1)
4(b)(ii)	a curved line from the origin	B1
	a line from the origin with a continuously decreasing positive gradient	B1